

# Rudra Mitra

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## Education

University of Bridgeport | Bridgeport, CT

Graduated December 2025

B.Sc. in Computer Science with minor in Mathematics

GPA: 3.73/4.0

## Professional Experience

Software Development Associate

May 2026 – Present

[20|20 Mobile Corp](#)

Bridgeport, CT

- Designing and developing software system ([Team Hub](#)) by using NetSuite that support applications and operations
- building full software stacks, including backend services, APIs, automation tools, and distributed systems.
- Participating in architecture and design of microservices, virtualization, SDN/NFV sustems, and big-data platforms  
Contributing to emerging technologies, proposing improvements and new frameworks

Lab Research Assistant

February 2025 – January 2026

Extreme Environment Engineering Lab, University of Bridgeport

Bridgeport, CT

- Engaged in **NASA-Connecticut space grant research** project by contributing to high-altitude ballooning missions and robotic payload development (HAM- High Altitude Monkey), and **Navy Grant** submersible research projects.
- Developed and integrated a radiation sensor system by updating and modifying software by using **Arduino-based code**, working with microcontrollers, utilizing **CAD models** for payload design, and improving data collection accuracy in extreme environments.
- Led STEM outreach via the universitys **STEM Bus**, inspiring and training school students in hands-on engineering, computer science, and AI projects during summer programs. [Link for STEM Bus](#)

STEM Tutor -Computer Science, Mathematics, and Physics

August 2023 – March 2025

[STEM Tutoring Center](#), University of Bridgeport

Bridgeport, CT

- Conducted individual tutoring sessions for over 240 students, diagnosed in learning challenges and assisted in mathematics (Calculus, Linear Algebra, Probability and Statics), physics, and computer science (Java, Python, Fundamentals Algorithm, Operating Systems, Discrete Structures) to enhance academic performance
- Collaborate with faculty to refine course resources and support through effective data management and assessment tracking.

## Publications

- **A Hybrid Deep Learning Model for Pneumonia Diagnosis: Bridging Gaps in Imaging and Patient History Data.**  
**Rudra Mitra**, Intiser Islam, Shariar Islam Saimon, Xingguo Xiong  
Published in 2025 ASEE Northeast Section Conference [Paper Link](#)
- **Free Vibration Characteristics of Single-Degree-of-Freedom (SDOF) Mechanical Systems: Investigating through Theory, Experimentation and Numerical Simulation** (In book: Engineering Research: Perspectives on Recent Advances Vol. 7, 4357))  
Abdullah Al Hossain Newaz, **Rudra Mitra**, Refat Jahan, Abdul Kadir [Book Chapter Link](#)
- **AI-Driven Temporal-Spatial Wildfire Monitoring: A Hybrid Approach Using Edge Computing and Cloud-Based Analytics.** (This research is supported by NASA Connecticut Space Grant Consortium under Faculty Research Grant, PTE Federal Award No.: 80NSSC20M0129, grant period: 7/1/2024-05/31/2025)  
**Rudra Mitra**, Intiser Islam, Shariar Islam Saimon, Xingguo Xiong, Navarun Gupta, Ahmed ElSayed  
Published in 2025 Northeast Section Conference [Link](#)
- **Introduction Comparing the Interpretability of XAI Techniques on Fake News Detection.**(This work is supported by the National Science Foundation(NSF), Hispanic Serving Institutions (HSI) Program, under grant agreement 2150048)  
Juan Rosario Medina, **Rudra Mitra**, Christopher Vazquez, Srilekha Jannabhatla  
Published in 2024 ASEE Northeast Section Conference [Link](#)
- **User-Defined Location Identity: Defining The Interactive Engagement Fabric of the Global Digital Ecosystem.**  
Omar Guzman, Aashrith Kanchala, John Louis, Joseph De Jesus, **Rudra Mitra**, Ausif Mahmood, Scott Kennedy  
Published in Published in 2025 ASEE Northeast Section Conference [Link](#)

- **Arduino-based Smart Fire Detection System Using Infrared and Gas Sensors** (funded by NASA Connecticut Space Grant Consortium under Faculty Research Grant, PTE Federal Award No.: 80NSSC20M0129, grant period: 7/1/2024-05/31/2025)  
Patricia Bermeo, **Rudra Mitra**, Shariar Islam Saimon  
Published in 2025 ASEE Northeast Section Conference [Link](#)

## Research Projects:

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- **Active Wildfire Burned Area Detection and Prediction:** Developed a deep learning wildfire analysis system using transformer- based models for wildfire detection, burned area segmentation, and progression prediction from VIIRS satellite imagery, achieving high precision ( $>0.79$ ) and recall (0.8360). Integrated multi-temporal data, adaptive thresholding, and ensemble methods for accurate fire mapping and predictive modeling with parameter-efficient architectures (510M). [Github](#)
- **IGVC Auto Navigation:** I contributed to an autonomous navigation project and participated in the IGVC 2025 (32nd Intelligent Ground Vehicle Competition, Rochester, Michigan), earned certification for my work. Collaborated with a team of PhD students at my university, I focused on the tracking system, lane detection, and automated speed control, implementing solutions using Ubuntu 22.04 on Linux. [Github](#)
- **NASA CT Space Grant Project:** I worked on a NASA-Connecticut space grant-based Nationwide Ballooning -Solar Maximum Radiation Project, where our team launches large scientific balloons carrying various payloads. My primary contribution involved in developing and deploying a radiation sensor to detect cosmic rays. I also worked on the HAM (High Altitude Monkey) project, a robotic monkey equipped with a camera and remote-control capabilities, which we successfully launched to high altitude for aerial observation and data collection. [Link](#), [News](#)

## Awards and Honors

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- Achieved a certificate for participating in IGVC 2025 (32nd Intelligent Ground Vehicle Competition, Rochester, Michigan) competition at Michigan, USA [Certificate](#)
- Elected to the Honor Society for Computing and Information Disciplines and achieved a certificate for Upsilon Pi Epsilon 2024 with a scholarship award for that. [Certificate](#)
- Earned UPE (Upsilon Pi Epsilon) 2025 scholarship award (1000 dollar) for academic excellence in computing and represented the University of Bridgeport at the national level. [Link](#), [Certificate](#)
- Received award for organizing the UB IEEE Hackathon 2026 at university of Bridgeport. [Link](#), [Certificate](#)

## Organization/Club

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### President of Math Club- University of Bridgeport

January 2024 – December 2025

University of Bridgeport

Bridgeport, CT

- Leading executive team members in executing strategic goals, developing budgets, and overseeing project timelines to ensure efficient completion.
- Organizing events aimed at promoting engagement within the campus community, utilizing problem-solving skills to manage logistics effectively, and assisting students with their courses, homework, and exam preparation.

### Treasurer and Event organizer of IEEE

September 2024 – January 2026

University of Bridgeport

Bridgeport, CT

- Organizing and promoting events ([IEEE Hackathon 2026 at University of Bridgeport](#)), workshops, and meetings, and following the timeline.
- Managing the funds and strengthening connections between academia and industry through initiatives that enhance skill in AI, data science and emerging computing technologies.
- Leading a weekly meeting to keep updated on our projects, integrating AI tools and software development, ensuring time and impactful outcomes, and generating or utilizing different strategies to achieve productive outcomes.

### Treasurer and Event organizer of Upsilon Pi Epsilon

November 2024 – January 2026

University of Bridgeport

Bridgeport, CT

- Overseeing chapter finances by supervising the collection of membership dues, tracking expenditures, and preparing detailed financial reports to support budgeting and compliance with UPE bylaws.
- Coordinating and executing student engagement events, including technical workshops and speaker sessions, boosting participation and enhancing the chapters visibility